

Complete Notes: High-Dimensional Geometry for Machine Learning

Definition

High-dimensional geometry studies data with many features (dimensions). Every feature represents one axis. ML datasets often have tens, hundreds, or thousands of dimensions.

Why It Matters

- Most ML datasets are high-dimensional.
- Helps understand distances, similarity, optimization, and learning behavior.
- Used in regression, classification, clustering, PCA, neural networks, and deep learning.

Loss Functions

Loss measures prediction error.

MAE (Mean Absolute Error)

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

Advantages: Easy to interpret, less affected by outliers.

Disadvantage: Not smooth at zero.

MSE (Mean Squared Error)

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Advantages: Smooth and differentiable, penalizes large errors more.

Disadvantage: Sensitive to outliers.

Gradient Descent

Goal: Minimize the loss function.

Update rule: $\beta_{\text{new}} = \beta_{\text{old}} - \alpha(\partial L / \partial \beta)$

β = model parameters (weights), α = learning rate, L = loss function.

Repeat until loss becomes minimum.

Partial Derivatives

$\partial y / \partial x_1$ means x_1 changes while all other variables stay constant.

Used because ML models have many variables.

Critical Points

At maxima/minima: $dy/dx = 0$.

Second derivative test:

$d^2y/dx^2 > 0 \rightarrow$ Local minimum.

$d^2y/dx^2 < 0 \rightarrow$ Local maximum.

$d^2y/dx^2 = 0 \rightarrow$ Test is inconclusive.

Challenges in High Dimensions

- Curse of dimensionality.
- Sparse data.
- Higher computation cost.
- Risk of overfitting.

Solutions

- Feature selection.
- Feature extraction (PCA).
- Regularization (L1/L2).
- Collect more quality data.

Exam Points

1. MAE is robust to outliers.
2. MSE heavily penalizes large errors.

3. Gradient descent uses derivatives to minimize loss.
4. Partial derivatives optimize one parameter at a time.
5. Local minimum: second derivative positive.
6. Local maximum: second derivative negative.